CAPSTONE CASE STUDY 4

Agriculture - HarvestLink Farmer Cooperative

Dataset: Capstone_4_Agriculture_Cooperative.xlsx

Business Context

HarvestLink is a farmer cooperative that supports smallholder producers across several districts. The cooperative wants to understand production, post-harvest losses, input costs, selling prices, and market-channel performance.

Key Challenges

- Yield and profitability vary significantly by crop and district.
- Post-harvest losses reduce the quantity available for sale.
- Farmers use different market channels with different selling prices.
- Input costs are increasing and management wants to identify the most profitable production areas.

Management Question

"How can we use data to improve farmer profitability, reduce post-harvest losses, and strengthen market decisions?"

Task Instructions for Learners

1. Analyse the Business Problem

- Analyse harvest quantity, quantity sold, losses, input cost, price, and estimated revenue.
- Compare crops, districts, market channels, and seasonal patterns.
- Calculate yield per acre, loss rate, revenue, and gross-margin indicators.
- Use crop and district lookup tables in the Power BI data model.

2. Provide Business Insights

- Which crops and districts generate the strongest margins?
- Where are post-harvest losses highest?
- Which market channels should the cooperative prioritise?

3. Develop a 10-Minute Presentation

Present a 10-minute cooperative performance briefing with a Power BI dashboard and recommendations for farmers and management.

4. Engage in a Q&A; Session

Defend your conclusions, DAX measures, visual choices, and recommendations using evidence from the data.

Minimum technical expectation: demonstrate Excel analysis, Power Query cleaning, a suitable data model, at least 6 DAX measures, interactive Power BI visuals/slicers, 5 insights, and 3 recommendations.